



PRE-BOARD EXAMINATION

FACULTY	:	COMPUTER SCIENCE & MULTIMEDIA
COURSE	:	BACHELOR OF INFORMATION TECHNOLOGY
YEAR / SEMESTER	:	SECOND YEAR / FOURTH SEMESTER
MODULE TITLE	:	ARTIFICIAL INTELLIGENCE
SUBJECT CODE	:	BIT 353
TIME ALLOWED	:	3 HOURS
DATE	:	27 th AUGUST
SET	:	B

Instruction to candidates

1. This question paper has THREE (3) Sections.
2. Answer **ALL** questions in Section A, MCQ.
3. Answer **5** questions in Section B, MSAQ.
4. Answer **2** questions in Section C, MEQ.
5. No scripts or answer sheets are to be taken out of the Examination Hall.
6. For Section A, answer in the OMR form provided.

Do not open this question paper until instructed

(Candidates are required to give their answers in their own words as far as practicable)

SECTION A

Multiple choice questions

(30*1=30)

1. **What is an essential characteristic of Artificial Intelligence (AI)?**
 - a) It requires human intervention at all times
 - b) It can process large amounts of data to make decisions
 - c) It always acts randomly
 - d) It only works with structured data
2. **Which of the following is a branch of AI?**
 - a) Database Management
 - b) Expert Systems
 - c) Network Security
 - d) Quantum Computing
3. **Which type of AI system has self-awareness and human-level intelligence?**
 - a) Narrow AI
 - b) General AI
 - c) Super AI
 - d) Machine Learning
4. **What is the main function of an intelligent agent?**
 - a) To act in a random manner
 - b) To maximize a performance measure by perceiving its environment and taking actions
 - c) To store large amounts of data without processing
 - d) To replace all human jobs
5. **Which AI agent learns from past experiences?**
 - a) Simple Reflex Agent
 - b) Model-Based Reflex Agent
 - c) Learning Agent
 - d) Utility-Based Agent
6. **What is the primary goal of search algorithms in AI?**
 - a) To find an optimal or near-optimal solution to a problem
 - b) To generate random solutions
 - c) To eliminate all possible solutions
 - d) To make problems more complex

7. **Which of the following is NOT a type of AI?**
 - a) Narrow AI (Weak AI)
 - b) General AI (Strong AI)
 - c) Super AI
 - d) Organic AI
8. **Which AI search algorithm guarantees finding the optimal solution if one exists?**
 - a) Depth-First Search (DFS)
 - b) Breadth-First Search (BFS)
 - c) Random Walk Algorithm
 - d) Hill Climbing
9. **What is the key feature of a rational agent?**
 - a) It ignores past experiences
 - b) It follows random actions
 - c) It maximizes performance based on its knowledge and environment
 - d) It does not interact with the environment
10. **Which of the following AI techniques is used for pattern recognition?**
 - a) Linear Regression
 - b) Expert Systems
 - c) Neural Networks
 - d) Game Theory
11. **What is the main function of a heuristic in AI?**
 - a) To increase computational complexity
 - b) To guide search algorithms efficiently towards the goal
 - c) To eliminate all possible solutions
 - d) To randomly choose actions
12. **Which type of learning algorithm requires labeled training data?**
 - a) Supervised Learning
 - b) Unsupervised Learning
 - c) Reinforcement Learning
 - d) Evolutionary Algorithms
13. **What does predicate logic add to propositional logic?**
 - a) Only Boolean operators like AND, OR, and NOT
 - b) Variables, functions, and quantifiers to express more complex statements
 - c) A method for solving optimization problems
 - d) A way to perform genetic mutations in AI

14. Which algorithm is best suited for solving classification problems?

- a) K-Means Clustering
- b) Linear Regression
- c) Decision Tree
- d) Apriori Algorithm

15. What is Deep Learning primarily based on?

- a) Decision Trees
- b) Artificial Neural Networks (ANNs)
- c) K-Nearest Neighbors (KNN)
- d) Support Vector Machines (SVM)

16. What is the role of an activation function in a neural network?

- a) It updates the weights of the neurons
- b) It determines the learning rate
- c) It introduces non-linearity to the model
- d) It reduces the number of layers in the network

17. Which of the following is a commonly used optimization algorithm in Deep Learning?

- a) K-Means
- b) Gradient Descent
- c) Apriori Algorithm
- d) Decision Tree

18. What is the purpose of the reward signal in reinforcement learning?

- a) To determine the initial state of the agent
- b) To guide the agent's learning by providing feedback on its actions
- c) To modify the environment based on the agent's actions
- d) To adjust the learning rate of the algorithm

19. Which deep learning architecture is best suited for processing sequential data like speech or text?

- a) Convolutional Neural Networks (CNNs)
- b) Recurrent Neural Networks (RNNs)
- c) Support Vector Machines (SVMs)
- d) Random Forest

20. What is the primary purpose of backpropagation in neural networks?

- a) To initialize the weights of the network
- b) To compute the loss function
- c) To adjust the weights of the network based on the error
- d) To calculate the accuracy of the model

21. What is a major ethical concern in AI?

- a) AI always makes unbiased decisions
- b) AI will never replace human jobs
- c) AI systems can reinforce societal biases if trained on biased data
- d) AI is always fair and transparent

22. In ethical AI, why is transparency important?

- a) To ensure AI models remain private
- b) To prevent public access to AI decision-making processes
- c) To allow users to understand how AI decisions are made
- d) To limit AI development

23. How can AI be used ethically in business?

- a) By replacing human workers without consequences
- b) By maximizing profits regardless of fairness
- c) By ensuring AI systems do not discriminate and are transparent
- d) By avoiding AI implementation completely

24. What is a key advantage of AI in healthcare?

- a) It can replace all medical professionals
- b) It can provide biased diagnoses
- c) It can analyze large datasets to improve diagnosis accuracy
- d) It eliminates the need for medical research

25. What is the purpose of AI in autonomous vehicles?

- a) To make human drivers unnecessary
- b) To provide real-time entertainment
- c) To assist in navigation and collision avoidance
- d) To improve fuel efficiency

26. What is the key challenge in AI adoption?

- a) Lack of interest
- b) Too many AI professionals
- c) Ethical and regulatory concerns
- d) AI is already fully developed

27. Which AI technique is commonly used in chatbots?

- a) Decision Trees
- b) Natural Language Processing (NLP)
- c) Genetic Algorithms
- d) Image Recognition

28. Which of the following is a key advantage of AI in finance?

- a) AI eliminates the need for human oversight in financial decisions
- b) AI always guarantees profit in stock trading
- c) AI helps detect fraudulent transactions through pattern recognition
- d) AI makes random financial predictions

29. What is the function of a knowledge-based system in AI?

- a) To store large amounts of unstructured data
- b) To make decisions without using prior knowledge
- c) To simulate human expert decision-making based on stored knowledge
- d) To replace all human experts in decision-making

30. Which of the following best describes the Turing Test?

- a) A test used to measure computer processing speed
- b) A method to check if an AI system has consciousness
- c) A test to determine if a machine can exhibit human-like intelligence in conversation
- d) A benchmark for evaluating deep learning models

SECTION B

Short Question Answer

Attempt any five (5) questions out of eight (8) questions

(5*6=30)

1. What is Artificial Intelligence (AI)? How does it differ from Machine Learning and Deep Learning?
2. Discuss about the four views of AI.
3. Discuss the components of a Problem Solving.
4. Explain the working mechanism of the Depth First Search (DFS) Method with a suitable example.
5. Differentiate between Deductive Reasoning and Inductive Reasoning.
6. Discuss the different stages of Data Preprocessing in Machine Learning.
7. Explain the components and phases of a Neural Network.
8. What are the ethical concerns related to the development of AI?

SECTION C
Long Question Answer

Attempt any two (2) questions. Q.3 (Case Study) is compulsory.

(2*20=40)

1. Define the K-Nearest Neighbors (K-NN) algorithm and explain its significance. A shop sells three types of products: electronics, clothes, and groceries. Based on a customer's age, gender, and purchase history, predict the type of product the customer is likely to purchase next. Use the K-NN algorithm to solve this classification problem, explaining the steps involved and the distance measure used.
2. Discuss the concept of Constraint Satisfaction Problems (CSPs) with examples. In a village, there are four people: two husbands and two wives. Each man can only travel with his wife, but cannot be alone with any other woman unless his wife is with him. The village has a river with a boat that holds only two people. How can all four people cross the river without breaking the rules?
3. **Case Study (Compulsory):** You are tasked with developing an AI system that can classify handwritten digits into 10 categories (0-9). The dataset contains images of handwritten digits in various styles and quality. Your goal is to design a machine learning model that can classify these images with high accuracy. Which type of machine learning model would you choose to perform this classification task and why? Explain the advantages of your selected model and any potential challenges you might face with the dataset. (e.g., Convolutional Neural Networks, Logistic Regression, etc.).

*****BEST OF LUCK*****